



Robot Chassis Design and Car Assembly

CAD Mechanical Design Portfolio

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Developed Using

Autodesk Fusion360

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1 PROJECT OVERVIEW

This project involved the design and development of a two-wheel differential drive robot chassis using Autodesk Fusion 360. The chassis serves as the structural foundation for mobile robotic systems and is intended to accommodate common robotics components including an Arduino UNO, motor driver, battery pack, DC gear motors, and a caster wheel.

The objective was to create a lightweight, manufacturable, and assembly-friendly platform that integrates mechanical and electronic components into a compact robotic system. The design emphasizes accessibility, stability, and ease of assembly while maintaining a professional appearance suitable for educational and prototyping applications.

2 DESIGN REQUIREMENTS

2.1 MECHANICAL REQUIREMENTS

- Overall chassis dimensions: 180 mm × 120 mm
- Plate thickness: 3 mm
- Rounded corners with 15 mm radius
- Lightweight yet structurally rigid design
- Adequate mounting provisions for all components

2.2 ELECTRONICS INTEGRATION

- The chassis must accommodate:
- Arduino UNO
- L298N Motor Driver
- Battery Pack
- Two DC Gear Motors
- Front Caster Wheel

2.3 DESIGN CONSIDERATIONS

- Balanced weight distribution
- Ease of maintenance
- Accessibility of USB and power connections
- Cable management
- Design for assembly principles

3 COMPONENT LAYOUT

The component arrangement was developed to achieve an optimal center of gravity and efficient utilization of space.

3.1 FRONT SECTION

The front portion contains the caster wheel mounting arrangement. The caster provides support while allowing unrestricted rotation during steering maneuvers.

3.2 CENTRAL SECTION

The Arduino UNO was positioned near the center of the chassis to simplify wiring and maintain balanced mass distribution. Mounting holes were designed according to standard Arduino dimensions.

3.3 ELECTRONICS SECTION

The L298N motor driver was placed between the Arduino and battery pack to minimize wiring lengths and improve accessibility.

3.4 REAR SECTION

The battery holder and DC gear motors were positioned toward the rear while maintaining overall balance. Motor mounting holes were created symmetrically to ensure proper wheel alignment.



Figure 1 Component Layout and Packaging Arrangement

4 CAD MODELLING

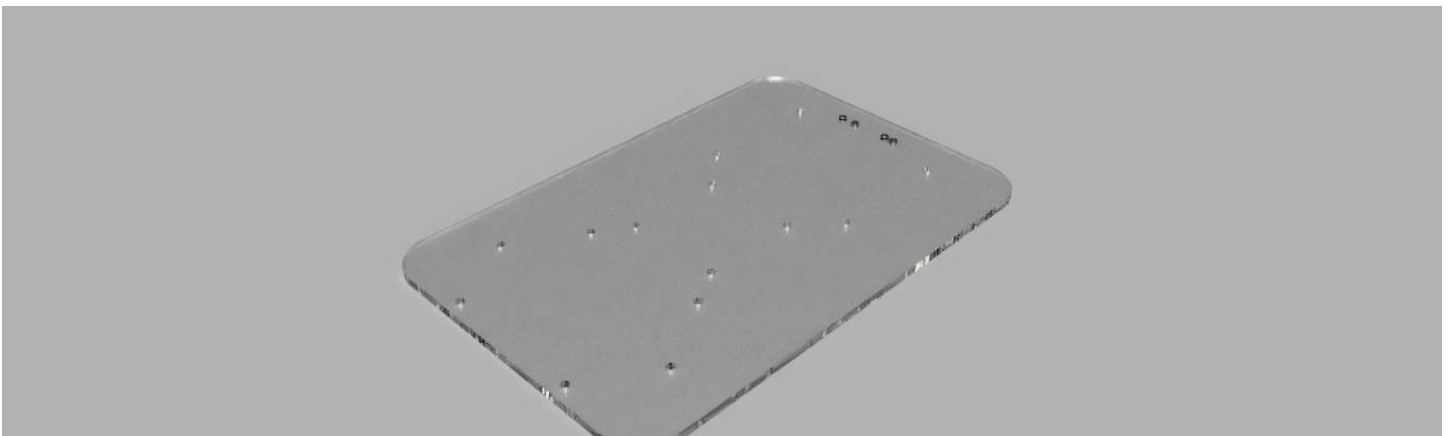


Figure 2 Chassis Base CAD Model

The entire chassis was modeled using Autodesk Fusion 360.

4.1 MODELING OPERATIONS USED

- Sketching
- Geometric Constraints
- Extrusions
- Extruded Cuts
- Circular Features
- Symmetry Operations
- Assembly Joints
- Material Assignment

4.2 MAJOR FEATURES

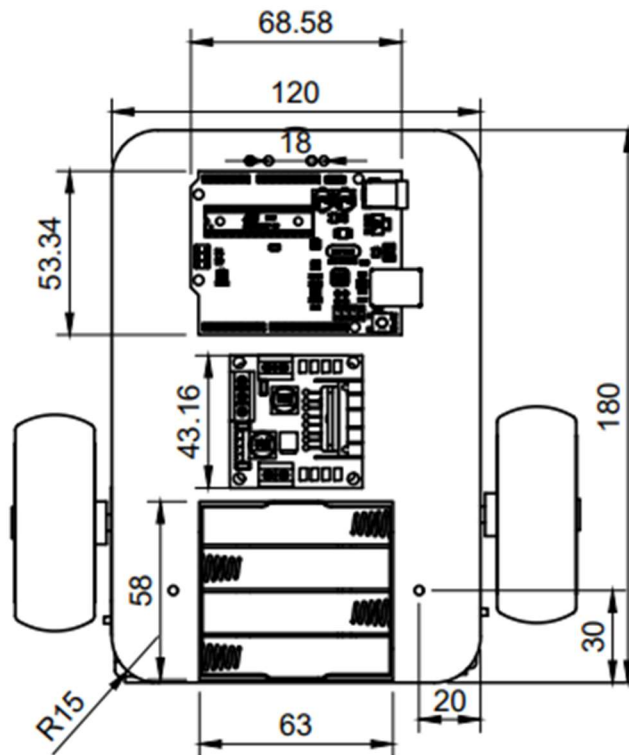


Figure 3 Major Design Features of the Chassis

4.2.1 Base Plate

- Length: 180 mm
- Width: 120 mm
- Thickness: 3 mm
- Corner Radius: 15 mm

4.2.2 Motor Mounts

- $\times \varnothing 3$ mm mounting holes
- Symmetrical placement
- Located 20 mm from side edges and 30 mm from rear edge

4.2.3 Arduino Mount

- Hole pattern: 68.6 mm $\times$ 53.4 mm
- 4 $\times \varnothing 3$ mm mounting holes

4.2.4 Motor Driver Mount

- Hole pattern: 36 mm × 36 mm
- 4 × Ø3 mm mounting holes

4.2.5 Battery Mount

- Reserved area: 60 mm × 60 mm
- Two integrated zip-tie slots

4.2.6 Caster Mount

- 2 × Ø3 mm mounting holes
- Hole spacing: 20 mm

5 ASSEMBLY CONSIDERATIONS

A simplified assembly was created to verify component placement and identify potential interference issues.

5.1 COMPONENTS INCLUDED

- Robot Chassis
- Arduino UNO
- L298N Motor Driver
- Battery Holder
- Two DC Gear Motors
- Two Drive Wheels
- Front Caster Wheel

5.2 INTERFERENCE ANALYSIS

The assembly was inspected to ensure:

- No overlapping components
- Adequate wheel clearance
- Accessible mounting hardware
- Proper spacing between electronic components

No major interference issues were observed during the assembly verification process.

6 ENGINEERING DIMENSIONS

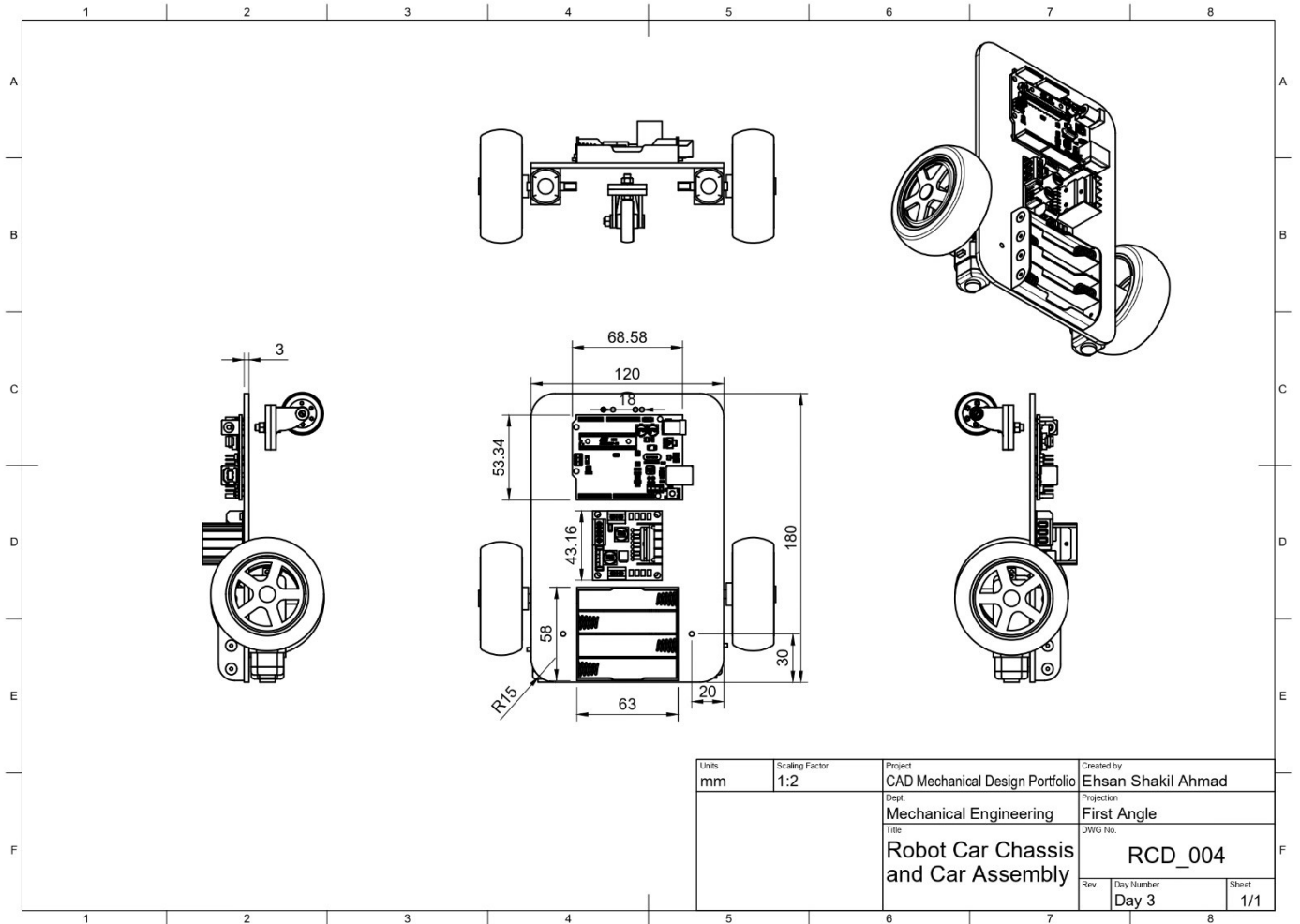


Figure 4 Engineering Drawing Package

Engineering drawings were produced according to ISO standards.

6.1 DRAWING FEATURES

- First-Angle Orthographic Projection
- Top View
- Front View
- Right Side View
- Isometric View

6.2 INCLUDED DIMENSIONS

- Overall chassis dimensions
- Hole diameters
- Hole locations
- Mounting patterns
- Cutout dimensions
- Material specifications

6.3 MANUFACTURING NOTES

- Material: Aluminum 6061-T6
- Thickness: 3 mm
- All dimensions in millimeters

7 RESULTS AND DESIGN EVALUATION

The robot chassis was successfully modeled and assembled in Autodesk Fusion 360, providing a complete proof-of-concept design for a two-wheel differential drive mobile robot.

The project achieved its primary objective of integrating mechanical and electronic components into a single chassis platform. Mounting provisions for the Arduino UNO, L298N motor driver, battery holder, DC gear motors, and caster wheel were incorporated into the design.

However, during assembly and evaluation, several design limitations were identified:

7.1 OBSERVED ISSUES



Figure 5 Identified Design Limitations

7.1.1 Caster Wheel Mount Alignment

The caster wheel mounting location was not perfectly aligned with the chassis centerline and wheel geometry. This could affect ground contact and overall robot stability during operation.

7.1.2 Chassis Size Constraints

The selected chassis dimensions proved to be relatively small for the intended component arrangement. Limited space resulted in packaging challenges and reduced flexibility for wiring and future upgrades.

7.1.3 Motor Mount Alignment

The motor mounting locations were not perfectly positioned with respect to the wheel assembly. Additional adjustments are required to ensure proper wheel alignment and symmetrical drivetrain geometry.

7.1.4 Electronics Mounting Accuracy

The mounting holes for the Arduino UNO, L298N motor driver, and battery holder require further refinement. Hole locations were approximated during the initial design stage and do not yet provide a fully optimized fit for the actual hardware.

7.2 LESSONS LEARNED

This project highlighted the importance of:

- Accurate component measurements before modeling
- Design verification through assembly checks
- Maintaining adequate clearances between components

- Iterative design refinement during product development

Despite these shortcomings, the design successfully established the overall layout and structural concept of the robot chassis.

The project serves as a strong foundation for future iterations and demonstrates the engineering design process from concept generation to evaluation.

8 FUTURE IMPROVEMENTS

Based on the evaluation of the current design, several improvements have been identified for future versions.

8.1 MECHANICAL IMPROVEMENTS

- Increase overall chassis dimensions to provide additional space for components and wiring.
- Reposition motor mounting locations to improve wheel alignment and drivetrain symmetry.
- Redesign the caster wheel mounting arrangement for improved stability and load distribution.
- Add dedicated cable-routing channels and wire management features.

8.2 ELECTRONICS INTEGRATION IMPROVEMENTS

- Re-measure physical hardware and update mounting hole locations accordingly.
- Create dedicated standoff features for Arduino and motor driver mounting.
- Improve battery holder retention and accessibility.
- Add mounting provisions for sensors and additional electronic modules.

8.3 STRUCTURAL IMPROVEMENTS

- Optimize cutout geometry to reduce weight while maintaining stiffness.
- Perform basic structural analysis to evaluate chassis rigidity.
- Investigate alternative materials and manufacturing methods.

8.4 DESIGN ITERATION GOALS

The next revision of the chassis will focus on:

1. Improved dimensional accuracy.
2. Better component packaging.
3. Enhanced manufacturability.
4. Improved assembly compatibility.
5. A more professional and production-ready design.

Although the current chassis is not a finalized product, it successfully demonstrates the design methodology, CAD modeling workflow, assembly development, and engineering evaluation process. The experience gained from identifying and analyzing these issues will contribute significantly to the development of future robotics projects.

9 CONCLUSION



Figure 6 Final Robot Chassis Concept

A two-wheel differential drive robot chassis was successfully designed and modeled using Autodesk Fusion 360. The project combined CAD modeling, assembly design, robotics integration, and engineering drawing generation to create a complete mechanical design solution.

The final chassis provides a robust platform for robotics applications such as line following, obstacle avoidance, autonomous navigation, and educational mechatronics projects. The project demonstrates practical mechanical engineering skills including design for assembly, component integration, CAD modeling, and engineering documentation.